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MODELS, NOT MAGIC

GENERALISATION, OVERFITTING, AND HOW ML MISLEADS

Week 3 · Semester 1, 2026

Current Advances in Psychological Methods & Analyses

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TODAY'S AGENDA

1. **Speaking ML** — translating stats vocabulary into ML language
2. **What is a model?** — demystifying the word
3. **Generalisation** — the fundamental goal of ML
4. **Train/test splits** — the simplest defence against self-deception
5. **Overfitting & underfitting** — when models lie to you
6. **Bias–variance trade-off** — the spectrum from too rigid to too flexible
7. **Cross-validation** — a more reliable way to evaluate
8. **Baselines & metrics** — how to know if your model actually works
9. **Regularisation preview** — setting up Week 4

THE BIG IDEA FOR THIS WEEK

> A model that fits your data perfectly is almost certainly **lying to you**.
>
> The real goal isn't to explain the data you have — it's to make **predictions about data you haven't seen yet**.

Understanding this distinction is what separates useful ML from self-deception.

SPEAKING ML

A translation guide from statistics to machine learning

STATS → ML: A TRANSLATION GUIDE

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What you know (stats)	What it's called in ML
IV / Predictor	**Feature**
DV / Outcome	**Target (regression) / Label (classification)**
Running a regression / fitting a model	**Training**
Your sample	**Training data**
Population you generalise to	**Test data (held-out)**
Coefficients / regression weights	**Model parameters / weights**
R^2, adjusted R^2	**Evaluation metrics (on test data)**

THE BIGGEST SHIFT

TRADITIONAL STATISTICS

Fit model to your **entire sample**

Ask: *"Is this effect real?"*

Assess significance (p-values, confidence intervals)

MACHINE LEARNING

Fit model to **part** of your data

Ask: *"Can I predict the rest?"*

Evaluate on held-out data (R^2, MAE)

Same data, fundamentally different question. Statistics asks *"is this real?"* — ML asks *"can I predict what happens next?"*

TWO FLAVOURS OF ML

SUPERVISED LEARNING

You have a **target variable** (labelled data)

- **Regression** — continuous target
  e.g., predicting depression score
- **Classification** — categorical target
  e.g., predicting diagnosis group

Weeks 3–6 of this course

UNSUPERVISED LEARNING

No target variable — looking for structure

- **Clustering** — finding natural groups
  e.g., patient subtypes
- **Dimensionality reduction** — simplifying
  e.g., PCA on questionnaire items

Weeks 7–8 of this course

OUR RUNNING EXAMPLE

Remember the **synthetic DASS dataset** from Week 2?

3,000 fake participants · lifestyle variables · DASS-21 depression, anxiety, stress scores

> **Features** (predictors): Sleep_hrs_night, Exercise_hrs_week, SocialMedia_hrs_week, ...
>
> **Target** (outcome): DASS_Depression (0–42 scale)
>
> This is **supervised learning → regression**

- Known correlations: Sleep r = −0.37, Exercise r = −0.27 with depression
- Students had highest depression (M = 20.3) vs employed full-time (M = 8.7)
- Can we formalise these patterns into a model that **predicts** depression for new people?

YOU'VE ALREADY DONE (A VERSION OF) ML

> If you've ever fitted a line of best fit or run a correlation, you've already done a version of machine learning.
>
> You just called it **statistics**.

The difference isn't the maths — it's the **question you ask**.

Stats: "Is sleep significantly related to depression?"

ML: "If I know someone sleeps 5 hours a night, what depression score do I predict?"

WHAT IS A MODEL, REALLY?

Demystifying the word

MODELS ARE EVERYWHERE

A **model** is a simplified version of reality that captures a pattern.

- "Sleep affects mood" — that's a model
- A regression equation — that's a model
- "Students are more stressed than working adults" — that's a model
- A decision tree — that's a model

> In **stats**, a model describes relationships in your sample.
>
> In **ML**, a model makes predictions about **new data** — people it has never seen before.

BEFORE A MODEL CAN LEARN: PREPROCESSING

Preprocessing = getting data ready for the model

- **Missing values** — you saw these in Week 2 (~2–3% of lifestyle columns)
  - Remove rows? Fill in with the mean? Depends on the situation.
- # **Encoding categories** — converting text to numbers
  - Gender (Male, Female, Non-binary) → numerical codes the model can use
- **Scaling** — putting variables on comparable ranges
  - Income in thousands vs. sleep in single digits — the model needs them comparable

We'll define these techniques properly in Week 4 when you put them into practice.

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GENERALISATION

The fundamental goal of machine learning

THE CORE GOAL: GENERALISATION

Generalisation = making accurate predictions about data the model has **never seen**.

- A model that explains your sample perfectly but fails on new participants is **worthless**
- We want models that capture **real patterns** — not noise specific to our particular sample
- This is where most ML projects go wrong — optimising for training data instead of new data

THE EXAM ANALOGY

MEMORISING

Memorise every past exam paper word-for-word

Perfect marks on *those specific papers*

New questions? **In trouble.**

= Overfitting

UNDERSTANDING

Understand the underlying concepts

Can answer questions you've never seen before

Adapts to new situations

= Generalising

Understanding the concepts (generalising) beats memorising the specifics (overfitting) every time.

YOU ALREADY KNOW ABOUT GENERALISATION

In inferential statistics, you generalise from a **sample** to a **population**.

That's what p-values and confidence intervals are for — asking whether patterns in your sample likely exist in the broader population.

> ML takes this further: instead of asking *"is this effect real?"* it asks *"can I predict what will happen for a new person?"*

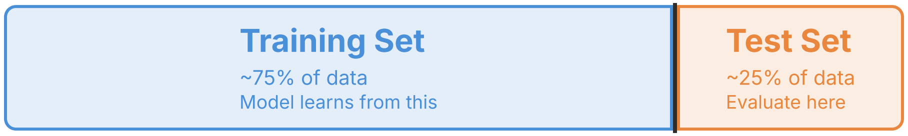
This connects directly to Week 1's theme: **prediction vs explanation** (Yarkoni & Westfall, 2017)

TRAIN / TEST SPLITS

The simplest defence against self-deception

SPLIT YOUR DATA BEFORE YOU START

Before building any model, split your data into two parts:



Like splitting your sample in half, running your analysis on one half, and checking whether the same pattern holds in the other — **before publishing**.

THE CARDINAL RULE

Never peek at the test set.

If you look at it — even once — during model development, it stops being "new" data and your evaluation is **contaminated**.

- Don't use test data to choose features
- Don't use test data to tune settings
- Don't use test data to compare models during development
- Only touch the test set **once, at the very end**, for your final evaluation

BEYOND THE BASIC SPLIT

Train/test is the foundation — but there are more sophisticated strategies:

- **Validation set** — a third split carved from training data, used to compare models and tune settings *without* touching the test set
  - Cross-validation (coming up next) achieves this without sacrificing data
- **Leave-One-Participant-Out (LOPO)** — hold out one person at a time, train on everyone else
  - Essential when you have repeated measures — prevents data from the same person leaking across splits
- **Leave-One-Group-Out (LOGO)** — hold out an entire site or group
  - Does a model trained at one hospital generalise to another?
- **Stratified splitting** — ensures categories (e.g., diagnosis, gender) are represented proportionally in each split

We'll explore these specialised methods in later weeks when we work with more complex datasets.

THINK ABOUT IT

A researcher builds 20 different models and picks the one that scores best on the test set. They report that model's test performance as their result.

Why is this a problem?

How is it similar to **p-hacking** in traditional statistics?

OVERFITTING & UNDERFITTING

When models lie to you

OVERFITTING

> The model **memorises** the training data — including its noise and quirks. Performs brilliantly on training data but poorly on anything new.

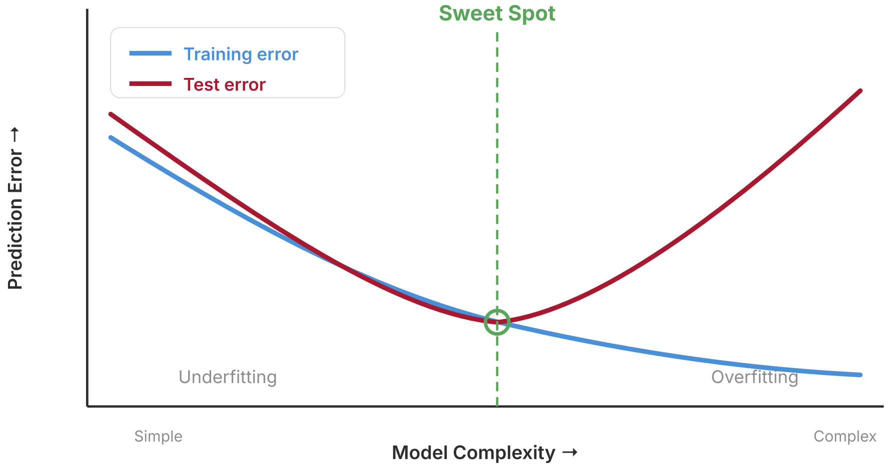
- Like the student who memorised past exam answers without understanding the material
- Training $R^2 = 0.95$ → "Amazing!"
  - Test $R^2 = 0.10$ → "Disaster."
- The model learned the **noise**, not the **signal**

UNDERFITTING

> The model is **too rigid** to capture real patterns. A straight line through clearly curved data.

- Like the student who only read the chapter summaries
- Training $R^2 = 0.05$ → "Hmm."
  - Test $R^2 = 0.04$ → "At least it's consistent... consistently bad."
- The model missed the **real patterns** in the data

THE SWEET SPOT



A CONCRETE EXAMPLE

Using the synthetic DASS dataset from Week 2:

Circular overfitting: If we include the individual DASS items (DASS_1, DASS_2, ... DASS_21) as features to predict DASS_Depression — the model "learns" the scoring formula. $R^2 \approx 1.0$ on training data. But it hasn't learned anything about how **lifestyle** relates to depression.

Clinical danger: A model appears to predict suicide risk with 95% accuracy in the training sample — but drops to 55% on new patients. A clinician relying on that model has **false confidence** in predictions barely better than a coin flip.

THINK ABOUT IT

Psychology traditionally uses relatively rigid models — t-tests, ANOVA, linear regression — that make strong assumptions.

ML offers far more flexible models.

When might a psychologist **prefer a biased but stable model** over a flexible but unstable one?

Think about: clinical decision-making, replication, and sample sizes.

THE BIAS–VARIANCE TRADE-OFF

The spectrum from too rigid to too flexible

BIAS

Bias = how far off the model's predictions are **on average**

- High bias → the model **consistently misses the truth**
  - A straight line fitted to a curved relationship
- High bias = **underfitting**
  - The model is too rigid to capture the real pattern
- Example: using only age to predict depression (ignoring sleep, exercise, social support...)
  - The model is systematically wrong because it's missing important information

VARIANCE

Variance = how much predictions **change** when trained on different samples

- High variance → the model is **unstable**
  - A wiggly curve that changes dramatically with each new dataset
- High variance = **overfitting**
  - The model is fitting noise, not signal
- Example: a complex model using all 44 variables on a sample of 50
  - Train it on a different 50 people and you get completely different results

THE TRADE-OFF

Reducing bias tends to **increase** variance, and vice versa. You can't minimise both at once.

High Bias	← The trade-off →	Low Bias
Low Variance		High Variance
Underfitting		Overfitting

Stats models you know:
t-tests, ANOVA, linear regression

High bias, low variance — strong assumptions, stable results, may miss patterns

Flexible ML models:
neural networks, random forests

Low bias, high variance — fewer assumptions, capture complexity, need more data

THE REPLICATION CRISIS CONNECTION

> Complex analyses on small samples = **high variance** = findings that don't replicate.

- Psychology's replication crisis is partly a **bias–variance problem**
  - Flexible analyses on small, noisy samples → results that look compelling but don't hold up
- ML makes this trade-off **explicit** rather than hiding it behind a single p-value
  - You can see the gap between training and test performance — that gap *is* the overfitting
- Cross-validation (coming up next) forces you to confront this directly

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CROSS-VALIDATION

A more reliable way to evaluate

THE PROBLEM WITH A SINGLE SPLIT

- A single train/test split gives you **one estimate** of performance
- What if you got a **lucky** split? Or an **unlucky** one?
  - Maybe the easy-to-predict participants all ended up in your test set
- Your result depends on **which observations** happened to land where
- We need something more robust...

K-FOLD CROSS-VALIDATION

1. Split your data into **K equal chunks** (typically 5 or 10)
2. Train on **K−1 chunks**, test on the remaining one
3. Repeat **K times**, rotating which chunk is held out
4. **Average** the K performance scores

Think of it like running your study **K times** with different random samples. Instead of one p-value, you get K estimates of performance — and you can see how much they vary.

5-FOLD CROSS-VALIDATION

Fold 1	Test	Train	Train	Train	Train	→ R^2_1
Fold 2	Train	Test	Train	Train	Train	→ R^2_2
Fold 3	Train	Train	Test	Train	Train	→ R^2_3
Fold 4	Train	Train	Train	Test	Train	→ R^2_4
Fold 5	Train	Train	Train	Train	Test	→ R^2_5

Final score = Average of R^2_1 through R^2_5

CROSS-VALIDATION IN PRACTICE

- **K = 5 or K = 10** are the most common choices
  - K = 5 is faster; K = 10 gives slightly better estimates
- If performance is **consistent** across folds → you can be more confident the model generalises
- If performance **swings wildly** → something may be wrong
  - Too few observations? Data not representative? Model too complex?
- **Important:** CV is for model *development* — comparing approaches, tuning settings
  - Still keep a final held-out test set you only touch at the very end

Reference: de Rooij & Weeda (2020) — "Cross-Validation: A Method Every Psychologist Should Know"

A NOTE ON RANDOM SEEDS

Data splits are **random** — different rows land in training vs test each time you run the code.

- **Random seed** = a number that makes randomness reproducible
    - Same seed → same split → same results every time
- **During development:** fix your seed so results are stable while you build and compare models
- **Before reporting:** try a few different seeds and check your conclusions still hold
    - If results change dramatically with a different seed → warning sign

> This extends beyond splitting. Models like neural networks start from **random initial weights** — fixing those seeds makes your entire analysis reproducible. We'll revisit this in later weeks.

```
random_state=42   ← doing something important!
```

BASELINES & METRICS

How to know if your model actually works

BASELINE MODELS: "AM I ACTUALLY LEARNING ANYTHING?"

Before building anything complex, ask: **what's the dumbest possible prediction?**

> **Mean prediction baseline:** Predict the average target value for everyone.
>
> In our synthetic DASS dataset, mean depression ≈ 14. If we predict 14 for every person, MAE ≈ 9 points.
>
> **If your fancy model can't beat this, it hasn't learned anything useful.**

- Baselines keep you honest
- R^2 = 0.15 sounds bad — until baseline is R^2 = 0.00 and theoretical max is ~0.30
    - Context matters enormously in behavioural science

THINK ABOUT IT

A research team reports that their ML model predicts therapy outcomes with **72% accuracy**.

Sounds impressive — but what if **70% of patients improve regardless** of treatment?

Their model barely beats random guessing. Why do you think **baseline comparisons are so rarely reported** in published ML papers?

EVALUATION METRICS FOR REGRESSION

R^2

`Coefficient of determination`

Proportion of variance explained

R^2 = 0.30 → model explains 30% of variation in depression scores

Familiar from stats — but now calculated on **test data**

MAE

`Mean Absolute Error`

Average prediction error, in same units as target

MAE = 4.2 → predictions off by ~4 points on the 0–42 DASS scale

Intuitive — easy to explain to non-technical audiences

RMSE

`Root Mean Squared Error`

Like MAE but penalises large errors more

RMSE > MAE → some predictions are very far off

Use when big mistakes are especially costly (e.g., clinical risk)

Rule of thumb: report at least **R^2** (relative performance) and **MAE** (practical accuracy). Classification metrics (accuracy, precision, recall) come in Week 5.

WHAT COUNTS AS "GOOD"?

It depends on the domain. In behavioural science:

- **R^2 = 0.10 – 0.15** → small but meaningful
  - Predicting individual behaviour from a few survey items? That's real signal.
- **R^2 = 0.20 – 0.35** → quite good for individual-level prediction
  - Human behaviour is inherently noisy — we're not predicting physics.
- **R^2 > 0.50** → check for data leakage or circular features
  - Suspiciously good? Make sure you're not accidentally cheating.

> We're predicting what real, complex people do — not the trajectory of a billiard ball. Context matters.

REGULARISATION

A brief preview for Week 4

FIGHTING OVERFITTING WITH REGULARISATION

Regularisation = adding a penalty that discourages the model from becoming too complex

RIDGE (L2)

- **Shrinks** all coefficients toward zero
- Keeps all features, but makes them smaller
- "Don't let any single predictor dominate"

LASSO (L1)

- Can shrink coefficients **all the way to zero**
- Effectively **drops** unimportant features
- "Which predictors can I remove without losing much?"

STATS CONNECTION: STEPWISE REGRESSION

You may have encountered **stepwise regression** — adding or removing predictors based on significance.

> **Lasso** does something similar but more principled — it simultaneously fits the model and selects features in a single step, rather than testing one predictor at a time.

- These are the models you'll **build in Week 4's lab**
- You'll compare: baseline → linear regression → Ridge → Lasso
- The AI will help you write the code — you need to understand **what** the models do and **why**

THINK ABOUT IT

If a Lasso regression drops a feature entirely (sets its coefficient to zero), does that mean the feature is truly **unrelated** to the outcome?

Or could it mean something else?

Think about what happens when two features are **highly correlated** with each other.

COMMON MISCONCEPTIONS

- **"Higher R^2 is always better"**
  - Not if it comes from overfitting. Training $R^2 = 0.95$ but test $R^2 = 0.10$ is a disaster.
- **"More features = better model"**
  - Adding irrelevant features introduces noise. Sometimes less is more.
- **"$R^2 = 0.25$ means my model is bad"**
  - In behavioural science, that's quite respectable for individual-level prediction.
- **"Cross-validation guarantees good performance"**
  - It gives better estimates, but can still mislead with structured or clustered data.

THE ML PIPELINE — PUTTING IT ALL TOGETHER

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Data (Features + Target) → **Preprocess** (Clean, Scale, Encode) → **Split** (Train / Test + CV) → **Train** (Fit Model) → **Evaluate** (R^2, MAE, RMSE)

This is the workflow you'll follow in every lab — **Week 4** will be your first time through it.

GETTING READY FOR WEEK 4

Next week you'll put all of this into practice:

- Build regression models on a **real DASS dataset**
  - 39,775 real survey responses — not synthetic data this time
- Use **scikit-learn** (`sklearn`) — the standard Python library for ML
  - Your AI assistant helps with the code — you focus on what and why
- Compare: **Baseline → Linear Regression → Ridge → Lasso**
  - Which model generalises best to held-out data?
- LLM skill focus: **debugging**
  - When code breaks, share the error with your AI and guide it to a fix

YOUR NEW SHARED LANGUAGE WITH AI

The vocabulary from this week becomes your tool for communicating with AI coding assistants:

"My **features** are TIPI Extraversion and Emotional Stability. My **target** is DASS Depression. I want to **train** a **Ridge regression** with **5-fold cross-validation** and report **MAE** and **R^2**."

The AI knows **exactly** what you mean. That's the power of speaking ML.

Before next week: Read the companion reading if you haven't already. Review the key terms. You don't need to memorise code — just the concepts.

Homework: Download the Week 4 dataset *before* class!

```
conda activate psyc4411-env
cd weeks/week-04-lab/data
python download_data.py
```

KEY REFERENCES

- **Yarkoni & Westfall (2017)** — Prediction vs explanation in psychology
    - doi.org/10.1177/1745691617693393 — Open access
- **de Rooij & Weeda (2020)** — Cross-validation for psychologists
    - doi.org/10.1177/2515245919898466 — Open access
- **Poldrack, Huckins & Varoquaux (2020)** — Best practices for prediction studies
    - doi.org/10.1001/jamapsychiatry.2019.3671 — Free via PMC
- **James et al. (2023)** — *Intro to Statistical Learning (Python)*, Ch 2 & 5
    - statlearning.com — Free PDF

Full reading list: readings.md

QUESTIONS?

Next week: Your First ML Pipeline — Regression with scikit-learn

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